National University of Computer and Emerging Sciences (FAST)

EE

Applied Calculus (EE) Final (Fall 2015) **Time allowed: 3 hours** **Max.Marks:100**

<u>Please read the following instructions carefully before attempting any of the questions:</u>

- **Attempt all questions, all questions carry equal marks.**
- **Do not ask any question about the contents of this examination from anyone.**
- **If you think that there is something wrong with any of the questions, attempt it to the best of your understanding.**
- **If you believe that some essential piece of information is missing, make an appropriate assumption and use it to solve the problem.**
- **Write all steps, missing steps may lead to deduction of marks.**

<u>Question.1:</u> Find the following limits

(a) $\lim_{x \to 1} \frac{x-1}{|x-1|}$

(b) $\lim_{x \to \infty} \frac{\sin 2x}{x}$

(c) Define *f(1)* in a way that extends $f(x) = \frac{x^3-1}{x^2-1}$ to be continuous at *x=1*.

<u>Question.2:</u> Sketch a graph of the function $g(x) = \frac{(x+1)^2}{1+x^2}$ using the following steps.

(a) Identify where the extrema of g occur.

(b) Find the intervals on which g is increasing and the intervals on which g is decreasing.

(c) Find where the graph of g is concave up and where it is concave down.

(d) Sketch the general shape of the graph for g.

<u>Question.3:</u>

(a) Find the area of the regions enclosed by the curves $x - y^2 = 0$ and $x + 2y^2 = 3$.

(b) Evaluate $\iint_R y\sin(xy)\,dA$, where$R = [1,2] \times [0,\pi]$.

<u>Question.4:</u>

(a) Solve the following improper integral

$$\int_0^1 x \ln x \, dx$$

(b) Is the following integral convergent or divergent? Give reason for your answer. (Use comparison theorem)

$$\int_1^\infty \frac{e^x}{x} dx$$

<u>Question.5:</u>

(a) Find a plane through $P_1(1,2,3), P_2(3,2,1)$ and perpendicular to the plane $4x - y + 2z = 7$.

(b) Solve the inequality, express the solution set as interval or union of intervals. Also, show the solution on the real line.

$$\left|\frac{2}{x} - 4\right| < 3$$